

AI and Machine Learning Guide

Introduction

Artificial Intelligence (AI) is a broad field focused on creating systems that perform tasks requiring human intelligence. Machine Learning (ML) is a subset of AI where models learn patterns from data instead of following explicit rules. Deep Learning is a further subset that uses neural networks with multiple layers.

Supervised Learning

Supervised learning relies on labeled datasets. Common algorithms include Linear Regression, Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, and Neural Networks. Applications include spam detection, credit scoring, medical diagnosis, and image classification.

Unsupervised Learning

Unsupervised learning discovers hidden structures without labels. Popular techniques include K-Means clustering, DBSCAN, PCA, and Hierarchical Clustering. These methods are widely used for customer segmentation, anomaly detection, and dimensionality reduction.

Evaluation Metrics

Classification models are evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices. Regression models use MAE, MSE, RMSE, and R-squared. Choosing the right metric depends on the business problem.

Generative AI

Large Language Models use transformer architectures trained on massive datasets. Retrieval-Augmented Generation (RAG) improves factual accuracy by retrieving relevant documents before generating answers.

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